BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE422 : Artificial Intelligence | Summer 2026
Assignment on Ethical and Green AI

As AI becomes more integrated into our daily lives, its ethical implications, such as fairness, bias, and privacy, profoundly impact society. For instance, in class I showed you some examples of deep but hidden stereotypes in LLM/VLM generated content from my own research, such as tech engineer = white male, novice learner = white female, calamity victim = black people, and so on. Now, if an AI system used for screening job applications is trained on historical data reflecting past biases, it may unfairly prioritize certain demographics for technical roles while neglecting some other demographics, thus reinforcing systemic inequality. Simultaneously, the massive computational power required to train AI models consumes significant energy, posing serious risks to the environment. For example, training a single large AI model can have a carbon footprint equivalent to the emissions of five cars over their entire lifetimes, including their manufacturing. Sam Altman famously said "Do not say 'please' or 'thank you' to ChatGPT, those take tens of millions of dollars in power use!" Probably a hyperbole but you get the point.

So, while trying to make our LLMs more powerful, we also have to take these two important factors – ethics and environmental side-effects – into consideration.

In this assignment you are assigned a case study from the book "Ethics of Artificial Intelligence: Case Studies and Options for Addressing Ethical Challenges" [find the book PDF **here**]. Your task is to read the case study and answer the following questions in your own words. Try to use at least TWO references while answering each question, the more the better. Make sure to cite the references properly.

- You can find the case study assigned to you at the end of this document.
- You can base your answers entirely from the book, but that will not lead to good marks. You SHOULD do some research on your own and use some external references or examples (e.g. journal or conference papers, news outlets, etc; DO NOT use social media, wikipedia, or reddit as references).
- **We will check plagiarism, so any sort of plagiarism will result in you getting a 0.**

Questions:

1) **[CO4]**
   How does the system mentioned/proposed in the case study affect the society/environment? Is it sustainable? Explain briefly in your own words and use references where necessary.
   - Talk about how the system affects the people involved directly.
   - Long term effects of the system being in place.
   - Refer articles/journals/papers tackling or mentioning similar situations

2) **[CO5]**
   Do you think the system mentioned/proposed in the case study has any ethical violations in terms of bias, transparency and accountability?
   - Is the system biased against a certain group of people?
   - Is it transparent, meaning does it share to the public how it is making its evaluations?
   - Is the system/company holding itself accountable for any violations?
   - Refer articles/journals/papers regarding any violations or similar situations.

3) **[CO6]**
   Are there any legal implications if the mentioned/proposed system is deployed in the real world?
   Assess the responsibilities of an engineer in ensuring safety, fairness and compliance with regulatory frameworks before deploying the system.
   - Are there any legal repercussions of the system being used? If yes, what might they be?
   - What aspects of the system should an engineer be aware of to ensure safety, fairness, and compliance with regulatory frameworks?
   - Refer articles/journals/papers mentioning similar aspects of a system and/or regulatory frameworks.

Assigned case study:

Use the **LAST DIGIT** of your student ID to find your assigned case. For example, if your ID is **23101036**, your assigned case study is **Case 6.** If you study a case study different from the one assigned to you, you will get 0.

Last digit	Case No	Topic
0	0	**Gender Bias in Recruitment Tools (Book page 10. PDF page 20)**
1	1	**Discriminatory Use of AI in Law Enforcement and Predictive Policing (Book page 12. PDF page 22)**
2	2	**Use of Personal Data by Authoritarian Regimes (Book page 27, PDF page 36)**
3	3	**Genetic Privacy (Book page 28, PDF page 37)**
4	4	**Biometric Surveillance (Book page 30, PDF page 39)**
5	5	**Data Appropriation (Book page 40, PDF page 48)**
6	6	**Monetisation of Health Data (Book page 41, PDF page 49)**

7	7	**Pushing Sales During “Prime Vulnerability Moments” (Book page 54, PDF page 62)**
8	8	**Fatal Crash Involving a Self-driving Car (Book page 65, PDF page 72)**
9	9	**Unfair Dismissal (Book page 81. PDF page 88)**